Dual-ligand fluorination

01 / Experiment

Use the bench sheet. Staff weigh/coarse-load chemicals, handle caps/labware and start Python locally; the robot measures liquids. Keep generated CSVs with their protocol.

Chemistry and design

Pyridine-2-sulfonyl fluoride → 2-fluoropyridine. Nominal per vial: 25 µmol SM, 1.5 µmol each ligand, 3 µmol Pd(COD)(DQ), 2.5 µmol naphthalene; 250 µL toluene, 95 °C, 500 rpm, 4 h under air. No base/external fluoride. Solvent addition defines nominal concentrations; solute displacement/contraction are accepted approximations.

R1: **48 pairs covering all 30 ligands**, official 24-consensus +24-Sobol design; L07 excluded. Use the fixed protocols and matching maps. R2: **22 pairs +L17 tBuBrettPhos at 12 mol% total ligand**; L03 at most four wells. No repeats. Higher yield identifies a preliminary candidate; batch effects, reproducibility and cooperativity need later matched controls.

Robot stock preparation

1. Weigh the amounts in the weighing list; no mass-entry spreadsheet is needed. Weigh ligands in the glovebox; keep bottles there. Pd: **4 A3/B3/C3, 22.8 mg each; A4, 21.6 mg** (≤90 mg; keep 10 mg). Naphthalene: **60 mg in 1 C2**; `IS_MASTER` empty. Under air, the robot adds solvents, mixes and prepares reaction/analytical stocks. Preparation is inside campaign time.
2. Use six labeled **Axygen 12-channel reservoirs** as listed in the weighing/loading map; carriers occupy slots 3/6 in batches. Verify actual geometry, solvent and cover compatibility, liquid recovery, mixing and offsets before live release. Follow generated carrier/raw-load CSVs.
3. Run `prepare_stocks.py`: **p300 single left, p1000 single right**; tips 5 /8-9 respectively. Carriers occupy 3/6 temporarily; working racks 1/4. The supplied horizontal 8 ×30 mL reservoir in 2 is **single-channel only**. Robot prepares analytical matrix/masters/working stocks too.
4. Cover/store reserve reservoirs and cap/store the prepared analytical rack. Leave working tubes in mapped positions. Change right pipette to **p300 eight-channel GEN2**, replace tips 8/9 with 300 µL racks using the local App installation/calibration workflow. Keep left p300 single.

Slot	Reaction-stage use
1 / 4	2 mL working tubes; max 1.95 mL. Ligands at CSV addresses; 500 mM IS **1 C2**, SM/IS **4 D2**, Pd **4 A3/B3/C3/A4**; prep acid **4 D3**. Retain RockPhos, SM/IS and all four Pd tubes.
2	New EMPTY 12-channel reservoir. Robot prepares workup fluids here during the hold; no staff premixing/refills.
3 / 10	Reaction block at pipetting / heater-shaker positions; manual moves. During holds, 3 temporarily takes raw solvents.
6	Calibration plate during R1; later a fresh final sample plate per round. Three compatible 96-well LC plates total.
5 / 8 / 9	Single / multi / multi tips. Keep 7 and 11 empty.

Confirm block/adapter/latch, offsets, <=45 mm cap clearance and liquid immersion with **1.5 mm bottom clearance**. Confirm LC plate/carrier/seal/needle depth and the requested full 1.7-min method/cycle, UV channel, RTs/LOQs. Remaining physical setup facts.

Run R1

Start R1 Python with **48 NEW EMPTY vials** in the block at 3. Robot doses **25 µL SM/IS +75 µL each ligand +75 µL Pd**. Retain SM/IS tube **4 D2**, capped between rounds: the robot fills it with **1.89 mL** from the Axygen bulk stock; no R2 refill. Expected remainder after R1 is **689 µL**; R2 requires **at least 680 µL**. Record any loss and use the actual preparation ledger.

Hand-cap all R1 vials; move/latch at App prompts. Run **95 °C /500 rpm /4 h**. During this hold the robot prepares **14 standards** and quench/diluent. Swap analytical rack into 4 and **uncap its source tubes**; raw reservoir into 3. Seal/move calibration plate 6→HPLC. **Reaction stays on 10.** Restore rack 4 and remove raw reservoir when prompted.

Start a 4 h external timer at each App hold-start message. Preparation/pauses count. Code cannot stop heat while App-paused. At deadline, stop heat/shaking through the App; record deviations and do not resume dosing. Logging tolerance does not establish chemical equivalence.

Workup and final LC plates

1. Move sealed block to 3; **cool vials before opening**. App separately checks module <=30 °C. **Uncover reservoir 2; keep vials capped while robot pre-fills the fresh LC plate in 6 with 980 µL diluent/well.** Then uncap; robot adds **500 µL 9:1 ACN/water +0.15% FA**. Nominal extract: 750 µL, 0.1% acid, 1.25 mL headspace. Neither fluid contains IS.
2. **Leave reaction vials OPEN**; shake **750 rpm /2 min**, heater off, at the locally confirmed splash-free setting. Return block to 3, let bubbles settle briefly and sample while open. No recap/reopen cycle before sampling.
3. Sample promptly after shaking; LC diluent is already in place. Robot uses a fresh tip, two slow same-well prewets, **21 µL aspiration at 1.5 mm**, 1 µL retained reverse surplus and 2 µL air gap. Dispense nominal **20 µL into prefilled LC wells**, noncontact; discard tip. No manual dilution/vial transfer.
4. Recap reaction vials. Move final LC plate **6→cooled shaker 10**, A1 orientation unchanged. Robot shakes **UNCOVERED, heater OFF, 750 rpm /2 min** at the locally confirmed splash-free speed. Reaction block remains in 3.
5. Move LC plate **10→HPLC**, apply compatible pierceable seal and inject directly. R1: **R1-LC01 A1-H6**, 48 samples. R2: **R2-LC01 A1-H2 plus A3-G3**, 23 samples. **H3 is diluent only; never inject.** Follow generated sequence/maps; reaction and LC coordinates differ.

Final sample: 1 mL; nominal maximum analyte **666.7 µM**, naphthalene **66.7 µM**, FA **0.1%**, toluene **0.667%**. Use the calibrated injection volume. Seal after final shaking to limit evaporation. There are four final-sample plate moves plus the calibration-plate handoff.

R2 and finish

After ALL 48 R1 results pass QC, select 22 pairs +L17 reference. Run `refill_stocks.py` **before R2 dosing**. Remove block/LC plate from 3/6; present ORIGINAL Axygen reservoirs by ID. **Remove reservoir covers and uncap required working ligand tubes for top-ups.** No new weighing or hand aliquoting. Retain RockPhos; **keep SM/IS and all four Pd tubes capped during refill**. Pd routing: R1 **18/18/12 doses** from A3/B3/C3; R2 **six C3 +17 A4**.

Load **23 NEW EMPTY vials** in the SAME block; replace reservoir 2 with a NEW EMPTY one. Uncap required stocks. R2 Python doses **25 µL SM/IS each, then catalysts**; reference receives **150 µL L17**. Hand-cap and start the reaction. During R2 hold the robot makes fresh workup fluids; no new calibration. Repeat workup and final LC analysis.

Cap stock reserves between stages. Robot mixes sources before first use/every eight transfers, without air gaps/blowouts. Same-stock dosing tips may be reused only for validated noncontact transfers. Analytical samples always use fresh tips. Mixing can disperse solids; it does not prove a settling suspension is uniform.

Pause for visible layers, blocked tips, persistent bubbles, air uptake, dripping or tip contact. Do not sample an arbitrary layer, guess a top-up or filter/decant without recording the effect. Reprepare an uncertain analytical dilution robotically and record the replacement; retain rejected measurements. No separate homogeneity study is planned.

Finish after all 23 final injections and quantitative analysis. Save robot JSON logs, raw LC/MS/method/sequence files, times and deviations under `local_data/`. Run `python scripts/hte-return-package.py`; send its ZIP privately. It retains code, preparation/source ledgers, inputs, results, Claude records and file hashes; credentials are excluded.

02 / Stocks and calibration

Staff weigh; the OT-2 measures liquids. See the weighing list and `bulk_carriers.csv`. Naphthalene charge: **60 mg**.

Reaction stocks

Stock	Preparation target at 100% assay	Per reaction
Naphthalene source	**60 mg solid in 1 C2**; robot adds ~936 µL toluene for **500 mM**.	Used to make stocks below
SM + naphthalene	**403 mg SM** in **BULK-01-3 A1**; add **500 µL IS +2 mL toluene** robotically, mix and load **1.89 mL in 4 D2**. Nominal **1 M SM /100 mM IS**.	**25 µL**
Each ligand	**20 mM toluene**; weighed charges/reserves in CSV. Robot prepares bulk, mixes and fills working tubes.	**75 µL each**; reference 150 µL L17
Pd(COD)(DQ)	Direct in **4 A3/B3/C3: 22.8 mg each; A4: 21.6 mg**. Robot adds ~1.50/~1.43 mL toluene for **40 mM**. Total **90 mg**, ~5.94 mL; **10 mg for scope**.	75 µL
RockPhos L03	**~800 µL /20 mM**, 7.50 mg. Prepared directly in **1 C1**; retain that tube. **10 mg available, max 4 R2 wells.**	75 µL

Retain SM/IS 4 D2, all Pd tubes and RockPhos 1 C1 for both rounds; no SM/Pd refill. See experiment and ledgers.

Robot-made workup fluids

Each hold: robot makes 10% aqueous FA, then fluids in reservoir 2. Coarse-load raw solvents per CSV into A1-H1 reservoir in temporary slot 3.

Fluid	Composition	Use
Quench	**9:1 ACN/water, 0.15% v/v FA**, no IS	500 µL; open vials, heater off, 750 rpm /2 min
LC diluent	**1:1 water/ACN, 0.1% v/v FA**, no IS	980 µL +20 µL extract

Robot-made authentic standards

Weigh separate FIT and CHECK charges per analyte using the weighing list. Weigh liquid product by difference. Robot makes **10 mM** masters (SM ~6.45 mg /4 mL; product ~5.83 mg /6 mL), matched matrix and **0.5/5 mM FIT dilutions**. Retain reserves.

Robot makes **8 mL /2 mM analytical IS** from **32 µL of 500 mM source**, crediting carrier toluene toward the matched matrix; loads **1 mL into analytical A3**. `IS_MASTER` starts empty; weigh only `IS_SOURCE`. Solid naphthalene: CAS 91-20-3; MW 128.17.

Calibration during hold one - 14 injections

Robot fills/pipette-mixes **CALIBRATION**: SM A1-G1, product A2-G2. Seal/move directly to LC during R1. Follow generated preparation and injection maps.

Analyte, µM	Role	Stock, mM	Analyte, µL	Matrix, µL
0	Fit blank	None	0	966.667
10	Fit	0.5	20	946.667
50	Fit	0.5	100	866.667
150	Fit	5	30	936.667
350	Fit	5	70	896.667
800	Fit	5	160	806.667
666.667	Independent check	10, CHECK	66.667	900.000

Add **33.333 µL IS per row**. Fit blank +five levels; exclude CHECK. Require **R^2 >=0.995**, **check error <=15%**, fit error **<=15%** (**<=20%** at 10 µM), measured separation/LOQs. Quantitative floor: max(LOQ, lowest nonzero calibrator). Python calculates yields.

03 / Local pipeline

Browser: experimental procedure. Commands below are for the pipeline operator.

Frozen R1: **48 pairs, all 30 ligands**. No paid R1 rerun. The fixed OT-2 files are ready for local App import after the operator's equipment check. Physical actions: bench sheet.

Install and configure

Python 3.12: `uv sync --locked --python 3.12 --extra sources --extra robot`; prefix commands with `uv run`. Create `local_data/`. Copy `hte_inputs/campaign.json` to `local_data/campaign.json`, `hte_inputs/calibration_template.csv` to `local_data/calibration-plan.csv`, and `anthropic.key.example` to `anthropic.key`. Keep any existing files. Add the Claude key; never send it to the robot/Git.

Use schema **6** and **fresh naphthalene calibration**; old mesitylene calibration is invalid. Start from the current configuration; discard old stock-preparation outputs. Configure method/channel, cycle and RTs/LOQs. Default `reject` stops missing-analyte results; `censor` reports validated bounds. **Axygen bulk reservoirs:** check the actual geometry, solvent/cover compatibility and recoverable residual volumes before live release. Concentrations use nominal solvent addition; record actual times.

Robot preparation, then R1

Use the weighing list and the published fixed stock preparation and R1 protocols. For the later R2 software, create a local preparation manifest with the confirmed configuration; **this generates files only, and does not run stock preparation again**. No mass-entry CSV is needed:

```
python -m hte.cli --config local_data/campaign.json prepare-stocks \
  --output local_data/stock-prep --live
python scripts/check-fixed-campaign.py \
  --preparation local_data/stock-prep/preparation_manifest.json
```

If the check fails, reconcile the actual stock recipe before R2. Run the published fixed protocols locally: p300 single left /p1000 single right for stock preparation, then change right to p300 multi and tips 8/9 to 300 µL through the App installation/calibration workflow. Preserve the matching manifest at its default path. Start R1 with **48 NEW EMPTY vials**. Retain **SM/IS tube 4 D2**, capped between rounds. Send the robot-prepared calibration plate to LC using its map. Import/calibrate; set `lc_calibration_validated=true` after success. Run the published R1 workup after cooling. Keep reservoir 2; use a fresh final LC plate.

R2 and finish

Complete actual injection maps. After ALL 48 results:

```
python scripts/hte-lc-results.py next \
  --dosing protocols/fixed-campaign/round1/dosing.csv \
  --exports local_data/r1-batch*.csv \
  --injections local_data/r1-map*.csv --output output/r1-decision
```

Calibration defaults to `local_data/calibration.csv`. QC precedes Claude: **387 untested pairs →22 +L17 reference**, L03 max four. **After interruption rerun the SAME command/output**; accepted scores are reused. Inspect uncertain API requests; a new folder may repeat charges. Run printed `refill_stocks.py` **FIRST** for ligand top-ups; retain SM/IS **4 D2** and all four Pd tubes without refilling. Fit **23 NEW EMPTY vials** and a NEW EMPTY reservoir 2; run the printed dosing protocol, then `workup.py` **after reaction/cooling**. Follow the printed LC sequence; copy and complete its injection map in `local_data/`. If unreleased, resolve the validation issue and use the R2 export command.

After all 23 final injections, use the printed **NEXT DOSING** path below (or `output/r2-live/dosing.csv` if exported separately). Replace the placeholder. This only analyzes results; no Claude:

```
python scripts/hte-lc-results.py final \
  --dosing "PASTE THE NEXT DOSING PATH HERE" \
  --exports local_data/r2-batch*.csv \
  --injections local_data/r2-map*.csv --output output/r2-final
```

Return records

Save robot logs, raw LC/MS files, maps and run notes under `local_data/`. Stop writers and run `python scripts/hte-return-package.py`; send `transfer/hte-results.zip` privately.